

Generative AI Professional

Duration – 3 days

Participant – Mixed Background

Day 1 Focus: Getting Started with GenAI

1. Program Kickoff and GenAI Landscape

- What generative AI is, where it is used, and why it matters for professionals.
- Key distinctions between AI, ML, DL, GenAI, and agentic AI.
- A quick tour of current use cases across business, operations, engineering, and knowledge work.

2. Cold-Start Hands-On: First Experiments

- A guided first interaction with a GenAI assistant.
- Simple prompt experiments for summarization, rewriting, ideation, and Q&A.
- How to ask better questions and compare output quality.

3. Python Refresher for the Rest of the Program

- Essential Python concepts needed for hands-on exercises.
- Variables, data structures, functions, loops, imports, and basic file handling.
- Short practice tasks to make sure everyone can follow later labs.

4. Working With Text Data

- Basic text preprocessing concepts.
- Reading, cleaning, splitting, and structuring text inputs.
- Preparing simple inputs for downstream GenAI tasks.

Day 2 Focus: Core Generative AI Tools and Workflows

5. How Modern GenAI Tools Work

- High-level view of prompts, models, tokens, context, and responses.
- What makes LLM-based tools different from traditional software.
- Practical limits: hallucinations, context window, and output variability.

6. ChatGPT for Professional Work

- Using ChatGPT for summarization, drafting, analysis, brainstorming, and structured outputs.
- Prompt patterns for better productivity.
- Turning everyday tasks into reusable prompt workflows.

7. GitHub Copilot and Coding Assistance

- Using Copilot as an assistant in coding and scripting tasks.
- Code completion, explanation, refactoring, and test generation.
- Good practices for reviewing and validating AI-generated code.

8. Practical Prompting Patterns

- Role, task, context, constraints, and output format.
- Few-shot prompting, stepwise prompting, and iterative refinement.
- Making prompts reusable across teams and use cases.

Day 3 Focus: Standard Use Cases, Advanced Topics, and Responsible Adoption

9. Standard GenAI Use Cases

- Summarization, classification, extraction, drafting, rewriting, and search assistance.
- Building simple workflows around common business needs.
- When GenAI is useful versus when rule-based automation is enough.

10. Intro to Agents and Agentic Thinking

- What agents are and how they differ from simple chat interactions.
- Core building blocks: goal, tools, memory, and action loops.
- Good starter examples such as assistants, triage bots, and workflow helpers.

11. MCP and MCP Servers

- What MCP is and why it matters for tool-augmented AI.
- How MCP servers extend AI beyond chat into files, apps, and connected services.
- Why IDE integration is changing the developer and knowledge-worker experience.

12. GenAI Inside the IDE

- Using VS Code and similar environments with AI-connected workflows.
- Practical examples of context-aware assistance, code navigation, and task support.
- How IDE-connected tools improve productivity without replacing developer judgment.

13. Ethics, Legal, and Way Forward

- Responsible use of GenAI: privacy, security, bias, and transparency.
- Legal and organizational concerns: copyright, data handling, and policy alignment.
- Suggested next steps for participants: safe experimentation, skill-building, and team adoption.

